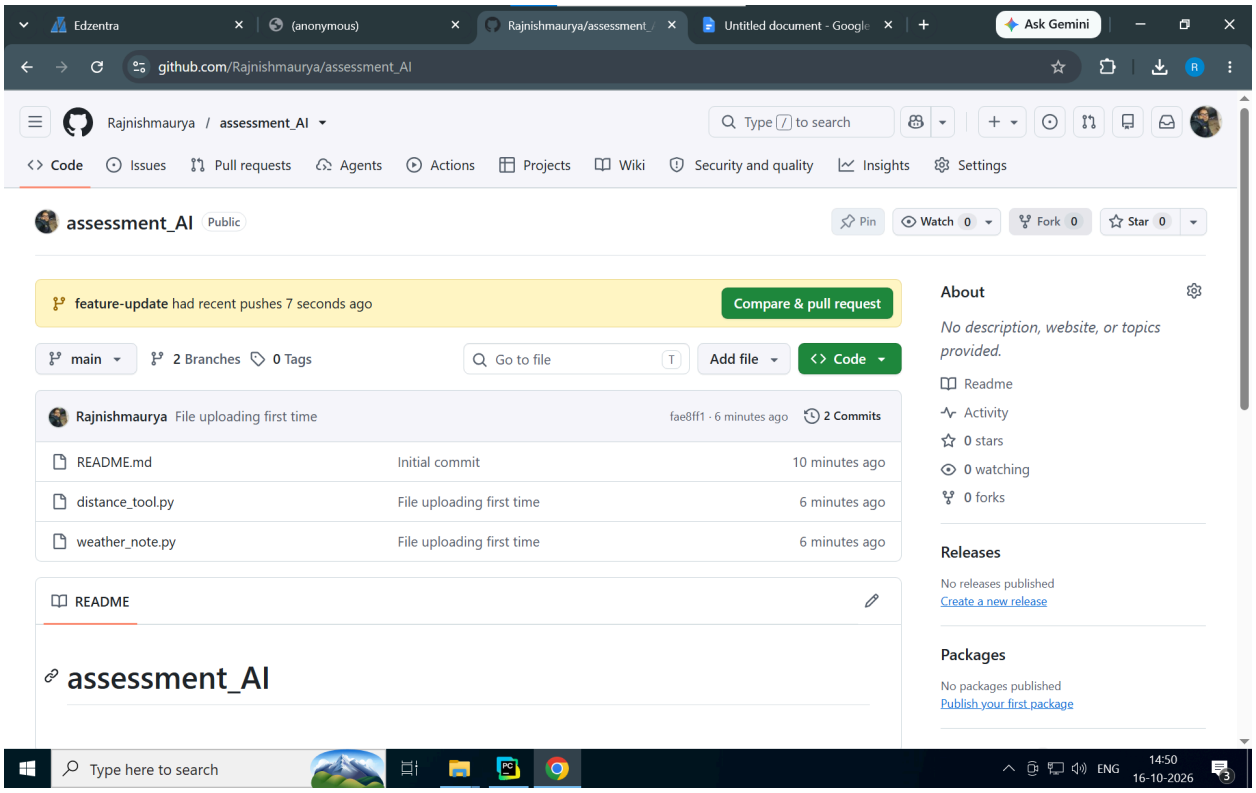
Section A : Short concept

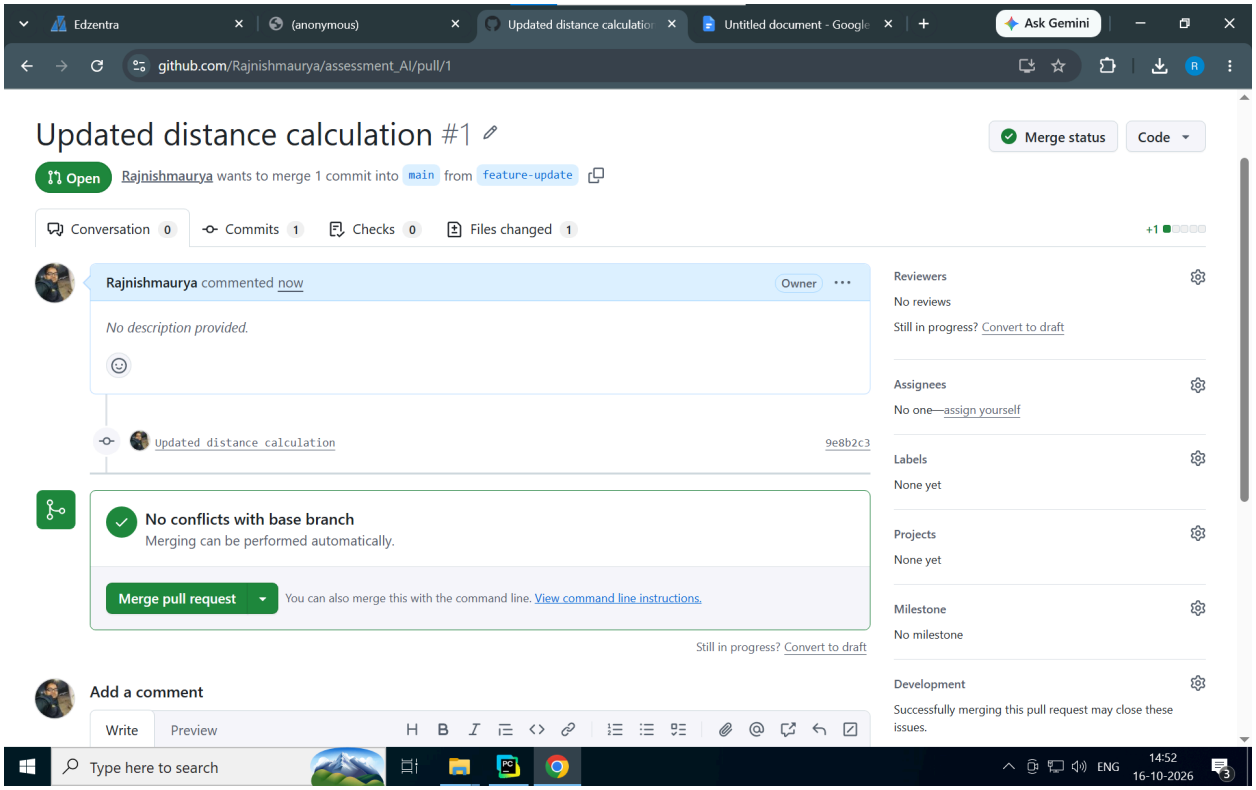
1. urban_mobility.csv - Structured

Reason: data is organized into rows and columns with fixed fields.

2. Equipment-events.json - Semistructured
   Reason: data uses json objects,keys,nested objects and arrays
3. Coastal_monitoring_log.txt - Unstructured
   reason : data consist of natural language sentences without a fixed tabular schema.

SECTION B Question 1 to 9





Answer these briefly:

B5. What is the difference between cloud infrastructure and on-premises infrastructure?

Cloud Infrastructure uses computing resources such as servers,storage and databases that are provided over the internet by cloud providers like Microsoft Azure. The company can scale resources as needed and opt for pays as you go method.

On-premises: The company owns or manages its physical servers and networking equipment as its own location. It is responsible for maintenance,upgrades,security and hardware costs.

B6. What is an Azure subscription?

An azure subscription is a logical boundary used to provision and manage azure resources. It helps organize resource usage,track billing, manage access and apply certain policies and limits.

B7. What is an Azure Resource Group?

An azure resource group is a logical container that holds azure resources such as virtual machines, databases, storage, accounts and networking components. These resources can be mapped together and access controls and resource related operations can be organized around the group.

B8. Arrange these in the correct hierarchy: Resource, Subscription, Resource Group.

Subscription-> Resource Group -> Resource

Subscription: Provide the management and billing.

Resource group: Organizes related resources with a subscription.

Resource: An individual Azure service such as virtual machines etc.

B9. Why might a company place related Azure resources in the same Resource Group?

Company places azure resources in the same resource group to simplify management, access control, monitoring, deployment and cost tracking.

SECTION C Question 15:
Normalize into separate tables:

suppliers(supplier_id,supplier_name,supplier_city)
parts(part_id,part_name,part_category)
purchase_orders(po_id,supplier_id,order_date,status)
purchase_order_items(po_id,part_id,quantity)

Repeated facts:
1. Supplier name and city repeat for every order from that supplier.
2. Part name and category repeat for every order containing that part.

Store supplier information once in suppliers and part information once in parts: foreign keys in purchase_orders and purchase_order_items.